
Symmetries in Physics — Exercise Sheet 3

Representations and equivalence

Variant: Questions only

Exercise 1. Let $C_3 = \langle r \rangle$ act on itself by left multiplication, and number the basis of its regular representation as $e_1 = e_e$, $e_2 = e_r$, $e_3 = e_{r^2}$.

- (i) Write the matrices of the operators $\rho(e)$, $\rho(r)$, $\rho(r^2)$ in this basis.
- (ii) Show that the representation is equivalent to a direct sum of one-dimensional representations. Exhibit the change of basis, and say which of the χ_k of Lecture 3 occur.

Exercise 2.

- (i) Find all one-dimensional representations of D_n for $n \geq 3$. How many are there when n is odd, and how many when n is even?
- (ii) Give the kernel of each.
- (iii) Show that none of them is faithful.

Exercise 3. Let D_4 act on the plane by its defining representation, with r the rotation through $\pi/2$ and $s(x_1, x_2) = (x_1, -x_2)$ the reflection in the x_1 -axis. Consider the function-space representation

$$(\rho(g)f)(\vec{r}) = f(g^{-1}\vec{r})$$

on the three-dimensional space of quadratic polynomials spanned by x_1^2 , x_1x_2 , x_2^2 .

- (i) Write the matrices of the operators $\rho(r)$ and $\rho(s)$ in this basis.
- (ii) Find a basis in which $\rho(r)$ and $\rho(s)$ are both diagonal. The representation is then a direct sum of three one-dimensional representations: say which ones of Exercise 2 they are.
- (iii) Which one-dimensional representation of D_4 does not occur?

Exercise 4. Lecture 3 writes the defining representation of D_3 in a basis in which s is the reflection in the y -axis. On Sheet 1, s was the reflection in the x -axis. With $R(\theta)$ the rotation through θ , the two versions are

$$D(r) = R(2\pi/3), \quad D(s) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad D'(r) = R(2\pi/3), \quad D'(s) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

- (i) Show that the quarter turn $T = R(\pi/2)$ satisfies $T D'(g) = D(g) T$ for $g = r$ and $g = s$, and explain why checking the two generators is enough. So the two versions are equivalent.
- (ii) Use Exercise 2 to show that the defining representation is not equivalent to any direct sum of two one-dimensional representations of D_3 .
- (iii) The rotations alone can be split: Lecture 3's last example makes every $D(r^k)$ diagonal in the basis $v_{\pm} = \begin{pmatrix} 1 \\ \mp i \end{pmatrix}$. Write $D(s)$ in that basis. What does the reflection do to the lines $\mathbb{C}v_+$ and $\mathbb{C}v_-$?

Exercise 5. Let $C_2 = \{e, p\}$ act on wave functions $\psi \in L^2(\mathbb{R})$ by the parity operator $(\Pi\psi)(x) = \psi(-x)$, where $\Pi = \rho(p)$.

- (i) Show that this is the function-space representation of the action $p \cdot x = -x$ of C_2 on $\mathbb{R}$.
- (ii) Show directly that Π maps its $+1$ and -1 eigenspaces to themselves, and that these are precisely the even and the odd functions.
- (iii) Prove that every ψ decomposes uniquely into one function of each kind, and state how p acts on the two pieces.